

DERBY, Australia

WMO: 952050

Lat: 17.3706S

Lon: 123.6611E

Elev: 7

StdP: 101.24

Time Zone: 8.00 (E08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	13.0	-9.3	1.7	25.8	-6.5	2.2	24.3	9.2	24.3	8.3	24.4	2.2	160	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	11.5	39.6	22.6	38.4	23.0	37.5	23.5	32.7	33.5	29.4	31.2	27.8	32.0	5.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
32.5	31.7	33.4	29.1	25.8	30.2	26.8	22.4	29.9	115.4	33.6	97.0	30.9	88.9	32.1	37.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.9	8.0	DB	8.5	42.7	1.7	1.2	7.2	43.6	6.3	44.3	5.3	45.0	4.1	45.9	(4)
(5)			WB	4.2	34.0	1.5	3.1	3.1	36.2	2.2	38.0	1.4	39.7	0.3	42.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	30.1	29.9	30.1	29.1	25.7	22.9	22.5	23.6	27.1	30.0	31.5	31.3	(6)
(7)		DBStd	3.77	1.97	1.90	1.72	1.89	2.36	2.55	2.29	2.11	2.10	1.63	1.43	1.90	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	2	0	0	0	0	0	1	1	0	0	0	0	0	(9)
(10)		CDD10.0	6502	623	556	622	573	488	388	389	422	513	620	646	661	(10)
(11)		CDD18.3	3462	365	323	363	323	229	139	132	164	263	362	396	403	(11)
(12)		CDH23.3	43506	4681	4077	4639	4050	2486	1409	1434	1911	3123	4638	5507	5552	(12)
(13)	CDH26.7	22824	2368	1999	2377	2157	1212	548	576	889	1697	2593	3234	3174	(13)	
(14)	Wind	WSAvg	4.0	4.5	4.2	3.6	3.3	3.5	3.7	3.5	3.7	3.9	4.5	4.8	4.9	(14)
(15)	Precipitation	PrecAvg	632	196	199	105	20	22	6	1	0	1	10	53		(15)
(16)		PrecMax	1102	618	511	370	79	140	67	13	1	10	10	41	194	(16)
(17)		PrecMin	262	18	21	2	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	195	154	108	90	22	40	13	3	0	2	2	13	50	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	38.8	38.9	38.5	36.7	33.8	33.6	35.5	39.0	40.8	41.6	41.2	(19)
(20)			MCWB	25.5	25.4	23.8	23.2	19.8	19.9	17.8	19.1	18.3	20.5	20.9	23.2	(20)
(21)		2%	DB	37.0	36.9	37.4	36.9	35.1	32.3	32.4	33.7	37.2	38.8	39.5	38.8	(21)
(22)			MCWB	26.3	26.4	24.3	22.8	19.8	18.9	17.4	18.4	19.0	20.9	22.3	24.7	(22)
(23)		5%	DB	35.8	35.6	36.2	35.8	33.9	31.2	31.3	32.5	35.9	37.3	38.1	37.3	(23)
(24)			MCWB	26.6	26.3	24.9	23.3	19.6	18.5	17.2	17.6	19.3	21.8	23.1	25.2	(24)
(25)		10%	DB	34.4	34.0	34.8	34.8	32.6	30.0	30.1	31.3	34.5	35.8	36.6	36.0	(25)
(26)			MCWB	26.5	26.5	25.3	23.0	19.4	18.1	17.0	16.8	19.5	22.3	23.6	25.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	33.0	30.1	34.7	26.6	29.3	23.6	32.0	33.3	33.5	32.7	34.9	(27)
(28)			MCDWB	34.7	33.2	30.9	35.4	29.6	29.5	26.7	32.2	33.9	34.5	33.9	35.9	(28)
(29)		2%	WB	28.8	28.5	28.1	28.4	24.7	24.2	21.2	24.1	26.1	26.8	27.8	29.2	(29)
(30)			MCDWB	31.7	32.5	32.3	31.2	28.7	26.3	27.1	25.1	28.4	31.1	31.5	30.7	(30)
(31)		5%	WB	27.6	27.7	27.4	26.5	23.5	22.0	19.9	21.0	23.6	25.6	26.6	27.3	(31)
(32)			MCDWB	32.2	32.2	31.8	30.8	28.5	27.1	26.8	26.4	28.0	30.9	31.9	32.4	(32)
(33)		10%	WB	27.0	27.1	26.9	25.8	22.5	20.6	18.8	19.9	22.5	25.0	26.0	26.7	(33)
(34)			MCDWB	31.9	31.8	31.3	30.2	28.0	26.2	26.5	26.6	28.4	30.3	31.5	32.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.1	9.4	11.9	13.7	14.3	15.3	16.3	15.6	12.9	11.5	9.8	(35)
(36)			MCDBR	10.3	10.4	11.5	12.8	14.8	14.7	16.4	17.1	17.1	15.0	13.3	11.7	(36)
(37)			MCWBR	3.9	3.9	4.3	5.4	5.8	6.2	6.6	7.1	6.7	6.1	5.1	4.4	(37)
(38)		5% WB	MCDWB	9.1	9.1	9.7	10.8	11.6	11.9	13.0	14.3	14.0	11.8	11.0	10.3	(38)
(39)			MCWBR	4.3	4.1	4.1	5.7	5.9	8.1	6.3	8.4	8.2	5.8	5.3	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.420	0.414	0.395	0.383	0.368	0.341	0.338	0.348	0.388	0.420	0.422	0.417	(40)	
(41)		taud	2.456	2.477	2.530	2.500	2.473	2.520	2.502	2.438	2.377	2.333	2.371	2.441	(41)	
(42)		Ebn at Noon	926	922	918	892	869	879	892	913	910	906	918	928	(42)	
(43)		Edh at Noon	121	117	107	104	99	91	95	108	123	134	131	122	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.37	6.42	6.13	5.96	5.16	4.88	5.16	6.03	6.91	7.38	7.49	6.94	(44)	
(45)		RadStd	0.58	0.68	0.67	0.24	0.33	0.27	0.27	0.17	0.15	0.23	0.26	0.43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.44	N/A	N/A	+0.54	+0.37	+0.32	N/A	N/A	+230	+231

Nomenclature: See separate page